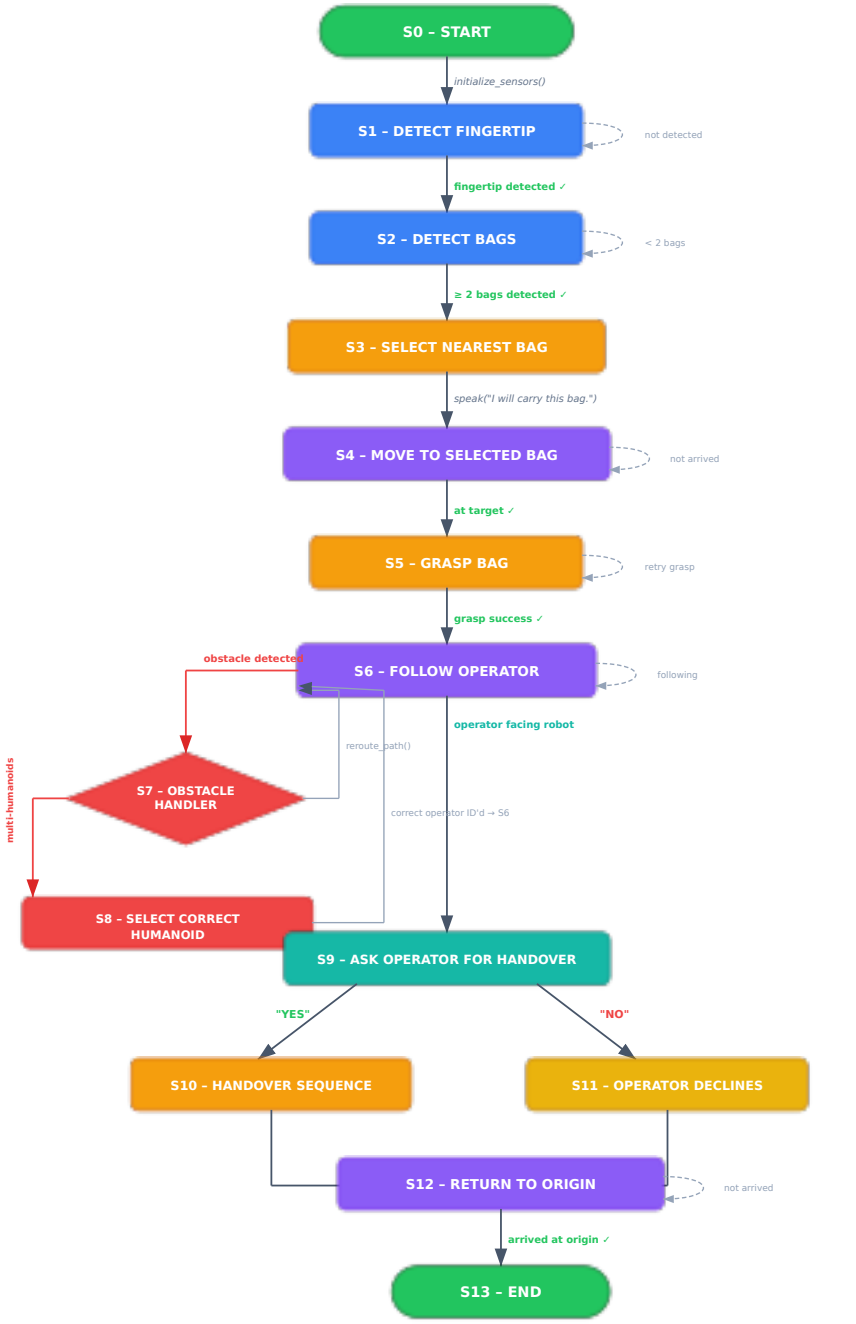
🤖 Carry My Luggage – Flowchart V4

Deterministic Reactive State Machine | PutrabotUPM Project | SMACC2 + ROS 2 Lifecycle

Start / End | Vision (VISION pkg) | Arm / Grasp (rx200_handbag_grasp, lift_bridge) | Navigation (my_robot_nav, putrabot_base) | Voice (voice pkg) | Obstacle / Re-ID (VISION + NAVIGATION)


S0 – START
initialize_sensors()
S1 – DETECT FINGERTIP
not detected
fingertip detected ✓
S2 – DETECT BAGS
< 2 bags
≥ 2 bags detected ✓
S3 – SELECT NEAREST BAG
speak("I will carry this bag.")
S4 – MOVE TO SELECTED BAG
not arrived
at target ✓
S5 – GRASP BAG
retry grasp
grasp success ✓
S6 – FOLLOW OPERATOR
following
obstacle detected
operator facing robot
reroute_path()
multi-humanoids
S7 – OBSTACLE HANDLER
correct operator ID'd → S6
S8 – SELECT CORRECT HUMANOID
S9 – ASK OPERATOR FOR HANDOVER
"YES"
"NO"
S10 – HANDOVER SEQUENCE
S11 – OPERATOR DECLINES
S12 – RETURN TO ORIGIN
not arrived
arrived at origin ✓
S13 – END

Package → State Mapping (src/ only)

CATEGORY	PACKAGE	STATES USED
VISION	VISION/ (vision_node.py, CamDetection.msg)	S1, S2, S3, S6, S7, S8
ARM/GRASP	rx200_handbag_grasp/ (grasp_node, yolo_handle_det.)	S3, S5, S10, S11
ARM/LIFT	lift_bridge/ (lift_serial_driver.py)	S5, S10
ARM/CORE	interbotix_ros_core/manipulators/toolboxes	S0, S5, S10
NAVIGATION	my_robot_nav/ (Nav2, return_home.py)	S4, S6, S7, S12
BASE	putrabot_base/ (cmd_vel, odometry, serial_bridge)	S4, S6, S12
VOICE	VOICE/ (text-to-speech + speech-to-text)	S3, S9, S11
SENSORS	realsense-ros, rplidar_ros, sllidar_ros2	S0 (bringup), S1–S12 (continuous)
DESCRIPTION	robot_description/ (URDF, TF), lift_description/	S0 (bringup), all states (TF tree)
MOVEIT	lift_moveit_config/, moveit_visual_tools/	S5, S10 (arm planning)

Note: INTEGRATION/ folder is currently empty — target folder for SMACC2 state machine package.
Hover over any state box for detailed package mapping information.